C 25599

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Name.....

Reg. No.....

# FOURTH SEMESTER B.Sc. DEGREE (SUPPLEMENTARY) EXAMINATION APRIL 2017

(UG–CCSS)

Mathematics

MM 4B 04—CALCULUS AND ANALYTIC GEOMETRY

Time : Three Hours

Maximum : 30 Weightage

I. Objective type questions. Answer *all* questions :

1 Find $y$ if $\ln y = 3t + 5$.

2 Evaluate $\int_0^{\pi/6} \tan 2x \, dx$.

3 Examine whether $\sum_{n=1}^{\infty} n^2$ converges or diverges.

4 Define Absolute convergence.

5 Find the Taylor polynomial of order zero generated by $f(x) = \frac{1}{x}$ at $a = 2$.

6 Find the directrix of the parabola $y^2 = 10x$.

7 Find the foci of the hyperbola $\frac{y^2}{4} - \frac{x^2}{5} = 1$.

8 Examine whether $3x^2 - 6xy + 3y^2 + 2x - 7 = 0$ represents a parabola, ellipse or hyperbola.

9 Write the parametric equations of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

10 Evaluate $\int_0^{\pi/2} e^{\sin x} \cos x \, dx$.

11 Evaluate $\frac{d}{dt}\left(\tanh\sqrt{1+t^2}\right)$.

12 State Leibniz's theorem for an alternating series.

(12 × ¼ = 3 weightage)

Turn over

II. Short answer type questions. Answer *all nine* questions :

13 Show that $x^2$ grows factor than $\ln x$ as $x \to \infty$.

14 Evaluate $\frac{d}{dx}\ln_{10}(3x+1)$.

15 Find $\frac{dy}{dx}$ if $y = x^x, x > 0$.

16 Examine whether the series :

$$5+\frac{2}{3}+1+\frac{1}{7}+\frac{1}{2}+\frac{1}{3!}+\frac{1}{4!}+\cdots+\frac{1}{k!}+\cdots \text{ converges.}$$

17 Prove that the alternating series :

$$1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots \text{converges.}$$

18 For what values of $x$ do the series $\sum_{n=1}^{\infty}(-1)^{n-1}\frac{x^n}{n}$ converges.

19 Find the Taylor series for $f(x) = e^x$ at $x = 0$.

20 Graph the set of points whose polar co-ordinates satisfy the conditions $1 \le r \le 2$ and $0 \le$

21 Find the Maclaurin series for $f(x) = \frac{1}{1+x}$.

III. Short essay questions. (Answer any *five* questions) : $(9 \times 1 = 9$ wei

22 Graph the curve $r^2 = 4\cos\theta$.

23 The standard parametrization of the circle of radius 1 centered at the point (0, 1) $xy$-plane is $x = \cos t, y = 1+\sin t, 0 \le t \le 2\pi$. Use this parametrization to find the the surface swept act by revolving the circle about the $x$-axis.

24 Find the tangent to the right-hand hyperbola branch :

$x = \sec t, y = \tan t, -\pi/2 < t < \frac{\pi}{2}$ at the point $(\sqrt{2},1)$, where $t = \pi/4$.

25 Find the Taylor polynomial generated by $f(x) = e^x$ at $x = 0$.

26 Prove that if $\sum_{n=1}^{\infty} |a_n|$ converges then $\sum_{n=1}^{\infty} a_n$ converges.

27 Show that $\lim_{x \to 0^+} (1+x)^{\frac{1}{x}} = e$.

28 Evaluate $\int_0^1 \sinh^2 x \, dx$.

(5 × 2 = 10 weightage)

IV. Essay questions. (Answer any *two* questions) :

29 Find the length of the Cardioid $r = 1 - \cos\theta$.

30 Show that the Maclaurin series for sin $x$ converges to sin $x$ for all $x$.

31 Using Integral test show that the $p$-series :

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{1^p} + \frac{1}{2^p} + \ldots + \frac{1}{n^p} + \cdots$$ converges if $p > 1$ and diverges if $p \leq 1$.

(2 × 4 = 8 weightage)